IXXN341,

Design For Health.



Enviromental Controller

Campbell Smith

Introduction

Based on my research in the first half of the project I focused on the application element of the enviromental controller.

Key issues with the current system are to do with the complexity, and time to set up a digital enviromental controller.

Whilst I can't address the time issue, due to how setting up IR sensors and recievers function in the existing hardware, I can address the complexity of the current app.

The focus of my design is the onboarding process, and ease of use once the device is set up. For my design I spent the most time on the onboarding, explaining how to set up all the buttons, and each buttons function works. Once the remotes are set up the usable remotes allow for users to change layouts, and display important information, such as mode and temperature for heatpump controls.

System Flow

General Flow

MND Patient recieves tablet

↓

Basic programs for communication and screen interface has been set up with technician and patient

↓

Patient has time with the device to develop skills for navigating

↓

Patient and technician discuss further needs and want

↓

Patient seeking enviromental controls is directed to enviro control software

Onboarding

Welcome Screen

↓

Description of Physical tasks required

↓

Singular button set up with diagram

↓

Testing button from app

↓

Second singular button set up

↓

Button test

↓

Full button set up opens

User can select any button they want to set up

↓

First click of button brings up a brief description of the button and tells user to press corresponding button on the remote

↓

User maps all the buttons they want

↓

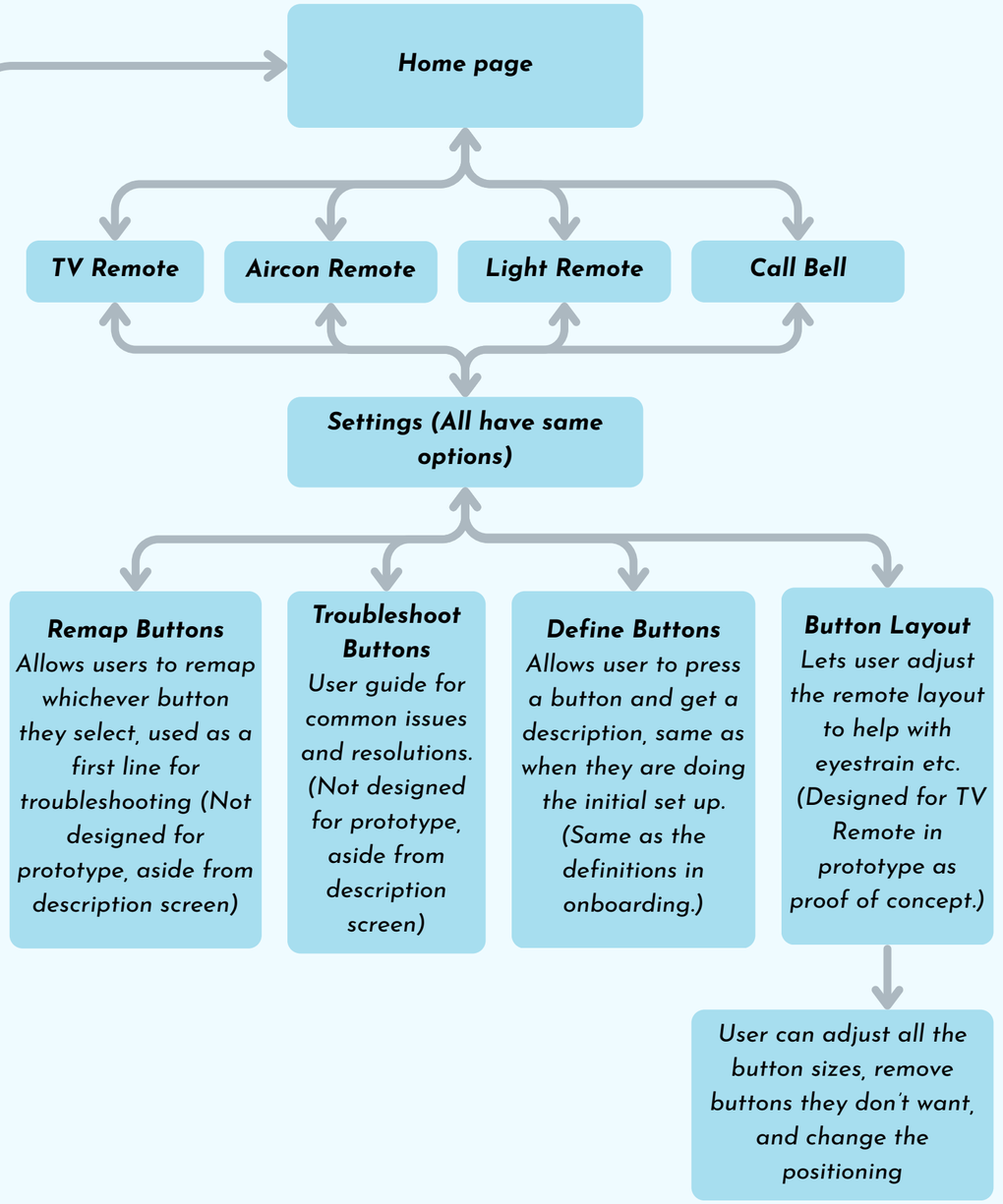
User presses the save button.

↓

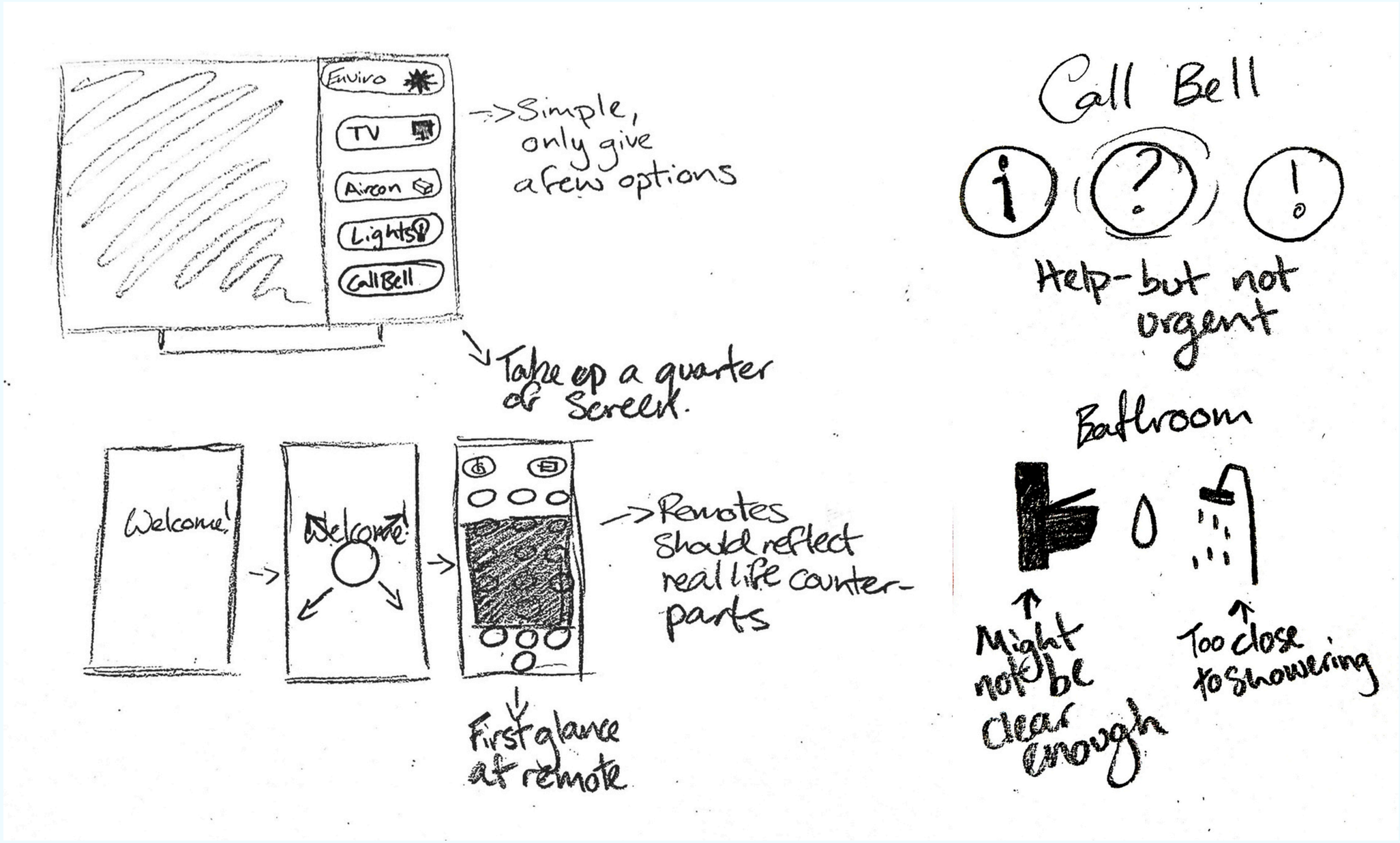
User goes to homescreen.

This onboarding process plays out in the full button set up option for every new remote the user wants to set up from the homescreen. (Not in prototype)

Main use flow

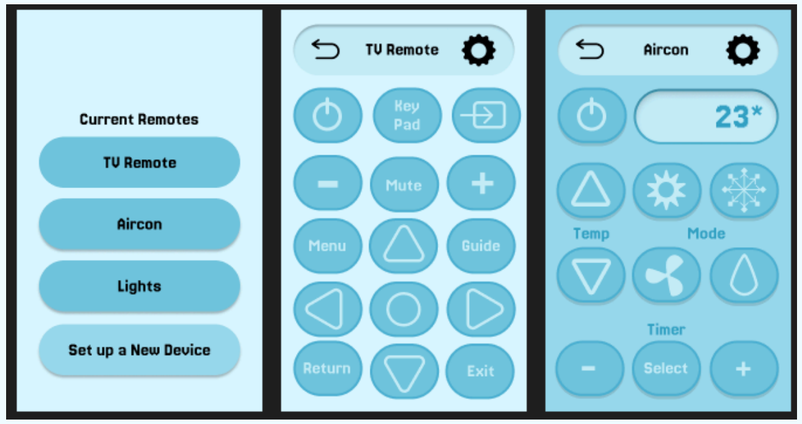


Sketches


Enviro
TV
Aircon
Lights
Call Bell
->Simple, only give a few options
Take up a quarter of Screen.
Welcome!
Welcome!
->Remotes should reflect real life counter-parts
First glance at remote
Call Bell
Help- but not urgent
Bathroom
Might not be clear enough
Too close to showering

Development

First Figma Design



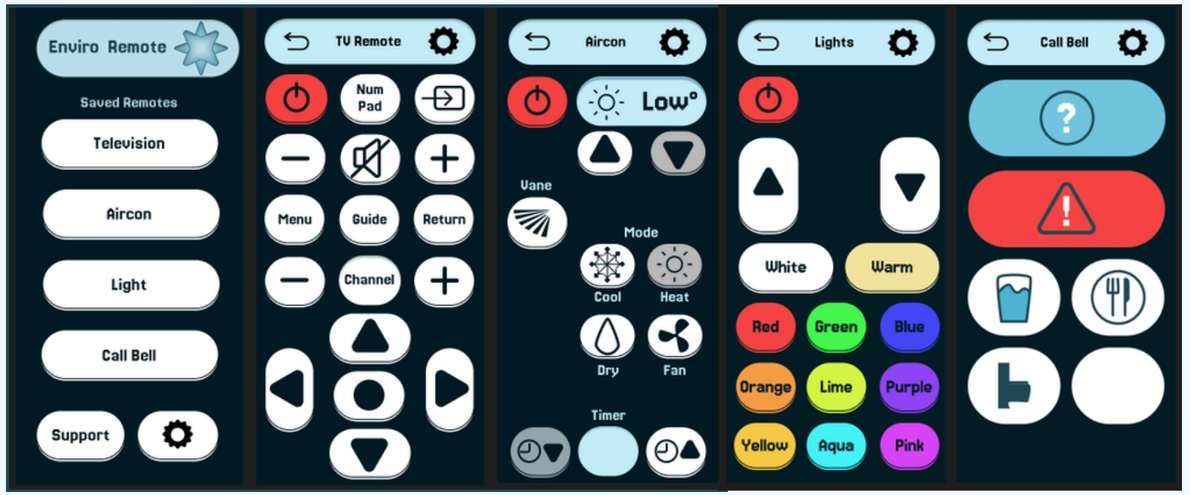
- Simple, missing key butttons, no variability.
- Colours aren't very high contrast, but was intial concept for looking at layout
- Small interface for allowing of multitasking

Second Round of Figma Design

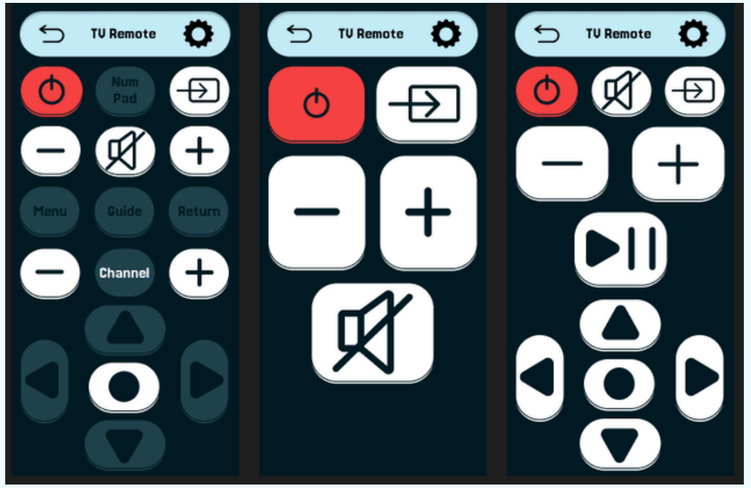


- Simple
- Higher contrast, colour blind friendly, but too bright
- Buttons non functioning without clear vision for what is a button and what is text
- Hard to look out for a user with eye strain.
- Final default Layout
- Long app so can be opened alongside other tabs and stay open whilst doing other things, allowing for more room in app.

Final Round of Figma Design



- Final colour pallette, high contrast with variability for non active buttons, and when buttons are pressed.
  - Still uses blues as a nod to the MND New Zealand branding.
- Added remotes for Lights, and a on tablet call bell, that could be connected to an IR Call bell or function by itself.
- Onboarding gets better wording here and works through the first two buttons one by one before opening to the full remote where the user chooses which buttons they want to map.
- Other Layout options shown, preset remotes for users, or can make their own variations to the remote layout.



AI Assisted testing

Personae #1, John Doe



- Age, 65,
- Male,
- MND Paitent,
- No Family History,
- Prior Occupation, Sales
- Family Connections,
  2 Children in 30s, Wife Jane Doe

Skills and Abilities

Strength

Dexterity

Constituion

Intelligence

Wisdom

Charisma

Technology

Characteristics,

- Lacks physical capabilities with limited movement,
- Witty, and Clever,
- Organised,
- Motivated by interests,
- Not technologically savy, but is capable in basic technology use due to work,

Goals,

- To be able to control basic household devices by self using his existing assistive technologies,
- Reduce workload on partner through indipendance,

Challenges,

- Not ICT Savy, so unable to devise a solution by self,
- Reliant on one Windows based device to use technology,
- Reliant on another person for any physical set up,
  - Experiencing mild cognitive decline,
    - Eye fatigue effects the longevity of technology use.

Personae #2, Jane Doe



- Age 62,
- Female,
- Occupation, Sales
- Family Connections,
  2 Children in 30s, Husband

Skills and Abilities

Strength

Dexterity

Constituion

Intelligence

Wisdom

Charisma

Technology

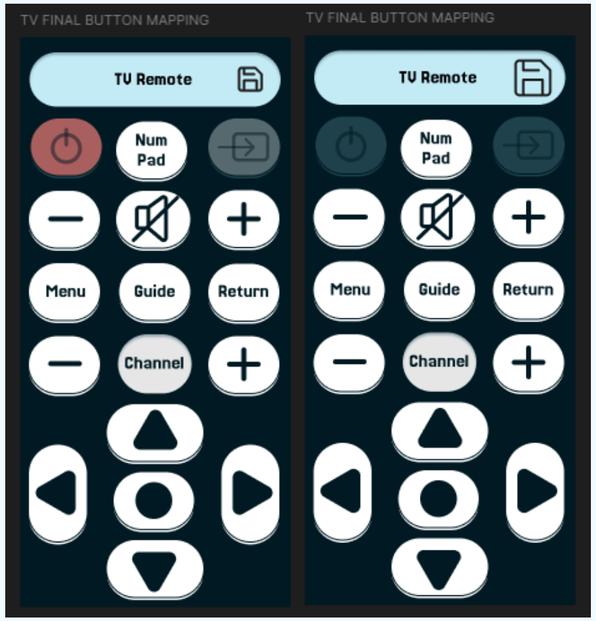
- Characteristics,
  - Patient,
  - Not the most ICT savy but is ready to ask children,
  - Dedicated to caring for husband,
- Goals,
  - Give as much autonomy to John as possible,
  - Reduce own workload for menial tasks to focus on bigger issues,
- Challenges,
  - Low Technology Confidence,
  - Reliant on others for technology advice,
  - Very busy time wise, and mentally taking care of John,

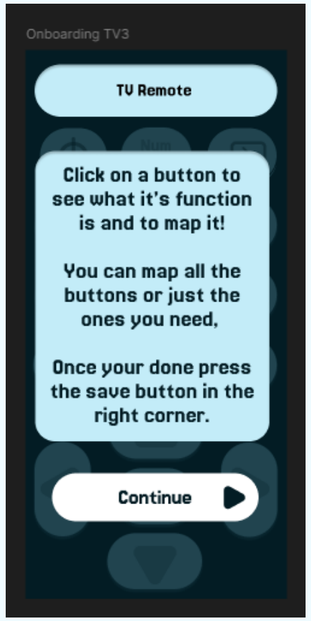
For the testing I used the same prompt twice, addressing the use case of the persona and his partner trying to set up a remote with only a few buttons that he wanted to use.

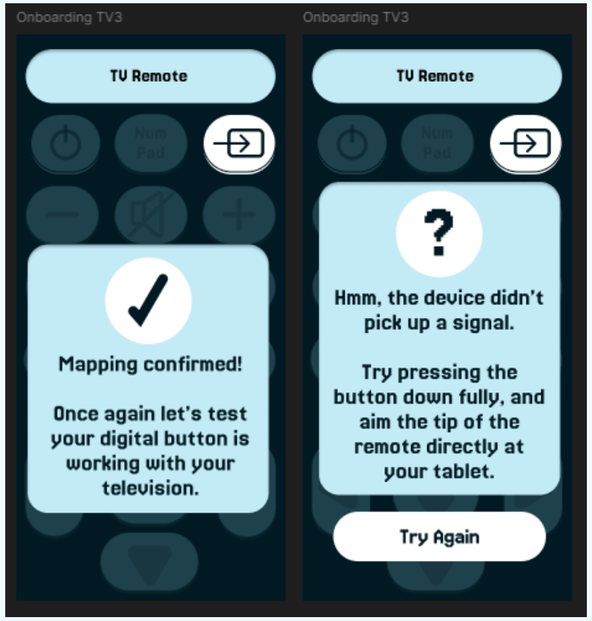
For the persona I reused my original patient persona, and caregiver/partner persona, representing the average users who would set this up.

Doing this I ignored the fact they may have assistance from a medical or technical person to assist them with setup due to in reality how often these patients aren't shown everything. Ideally this app should allow the users to set up themselves without intervention to reduce strain to medical resources, or it is such a simple design that a technician could do it in a far shorter time.

AI Assisted Development





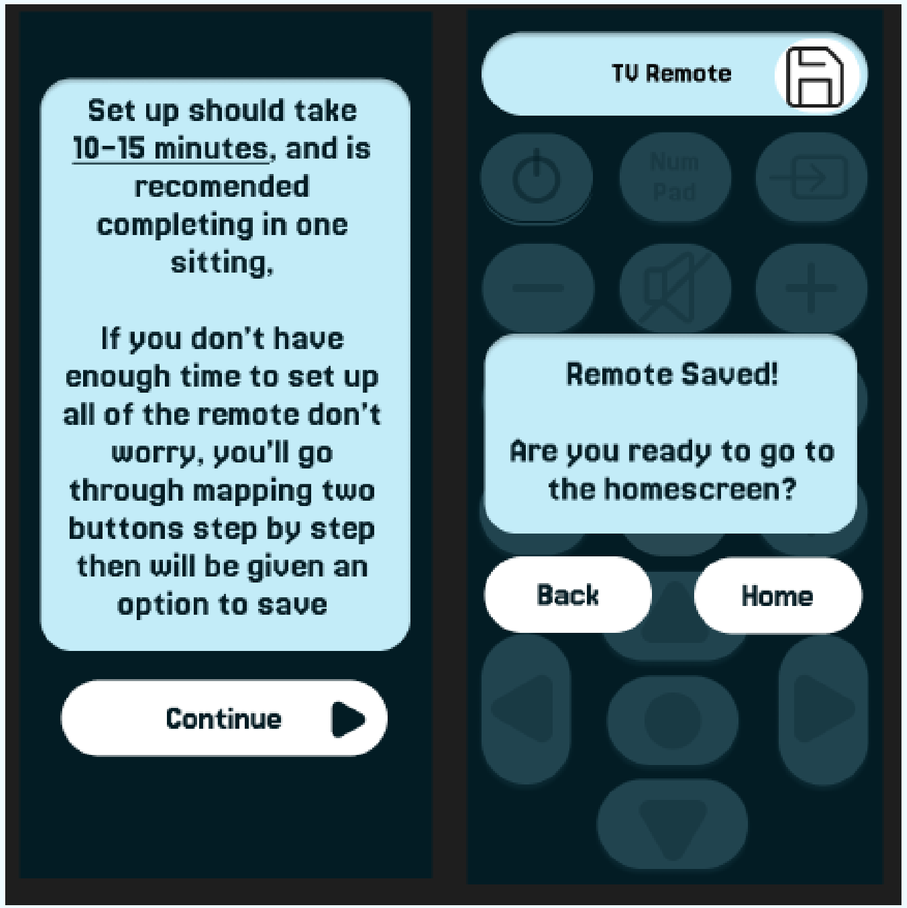




Prompt for simulation: *Simulation senario, the software has been added to John's tablet, he's already had the device a week or two so has adjusted somewhat to using his eyes to control the device, he wants to set up the software so he can control the tv in his room without disturbing Jane everytime he wants to turn the volume down or change channels, the time is roughly midday as this is when Jane is the most awake, John likely is more tired than Jane as tiredness is a common symptom of MND. (Keep in mind this app runs through figma so often there are buttons he presses that bring up correlating variants, or animations that play out that may appear as repeated frames)*

Friction Point	Decision	Reasoning	Response
Clear Onboarding Instructions	Adapt	Some of the language was flagged as not clear for the user profiles	Changing some phrasing such as 'Click' to 'Select'
Physical Remote to app feedback	Adapt	No pathway for failed mapping, no visual communcation the mapping is being programed.	Added text for the mapping screen that it may take a moment, fail screen with common issues pops up for simulated fail.
Colour Consistency	Accept	Greyed out icons for unactive buttons functions but is hard to look at	All unactive buttons have the same colour regardless of the buttons actual colour, making avaliable buttons highly visible.
Scope, Clear acknowledgement when the remote screen opens they don't need to map everything, just those are all the options.	Adapt	Due to limitations figma knowledge the concept of a flexi layout is simulated, most of the response was around the onboarding and later on when using own remote a number of the buttons being greyedout on the default.	Change of wording at the end of the onboarding to remind the user they only have to program what they need.

AI Assisted Development



Prompt for simulation: *Simulation senario, the software has been added to John's tablet, he's already had the device a week or two so has adjusted somewhat to using his eyes to control the device, he wants to set up the software so he can control the tv in his room without disturbing Jane everytime he wants to turn the volume down or change channels, the time is roughly midday as this is when Jane is the most awake, John likely is more tired than Jane as tiredness is a common symptom of MND. (Keep in mind this app runs through figma so often there are buttons he presses that bring up correlating variants, or animations that play out that may appear as repeated frames)*

Friction Point	Decision	Reasoning	Response
Specific Instructions to caregiver and user	Reject	AI claims requires further communication between the two to determine whiich action is taken by either or user	Not changing this language as a mobile MND patient may be programming the buttons themself and the actions are already self explanatory.
Physical Remote to app feedback	Reject (For Now)	Reassuring text was added but no further visual	No change for now, concerns about computer load when it comes to displaying mapping status and programming the IR signal in the background.
Save button not highllighted clearly	Accept	Confusing it is just a symbol on the task bar and not a button itself	Button added to task bar for the save icon
Allow Pausing and resumming	Adapt	Partial mappping is avaliable but no direct pause if user decides to come back to it later	Add a estimated set up time for the user to allocate time for set up, onboarding should be done in one sitting for the best understanding from the user.

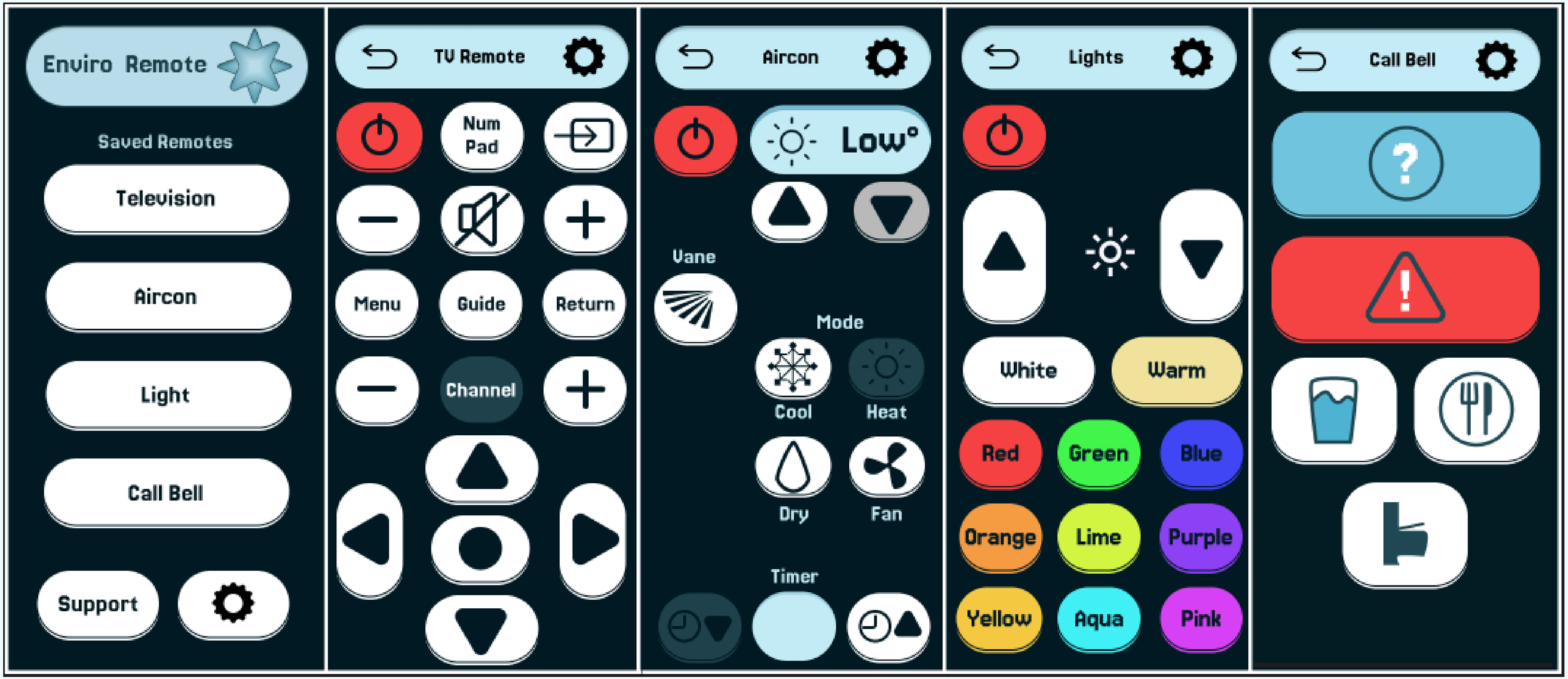
Reflection

Once I had an idea of focus I looked at building out a step by step onboarding, intially I ran it in three sections before opening to the whole remote, but ended up reducing it to two choosing Power and Source for their location on the remote and their common usecase.

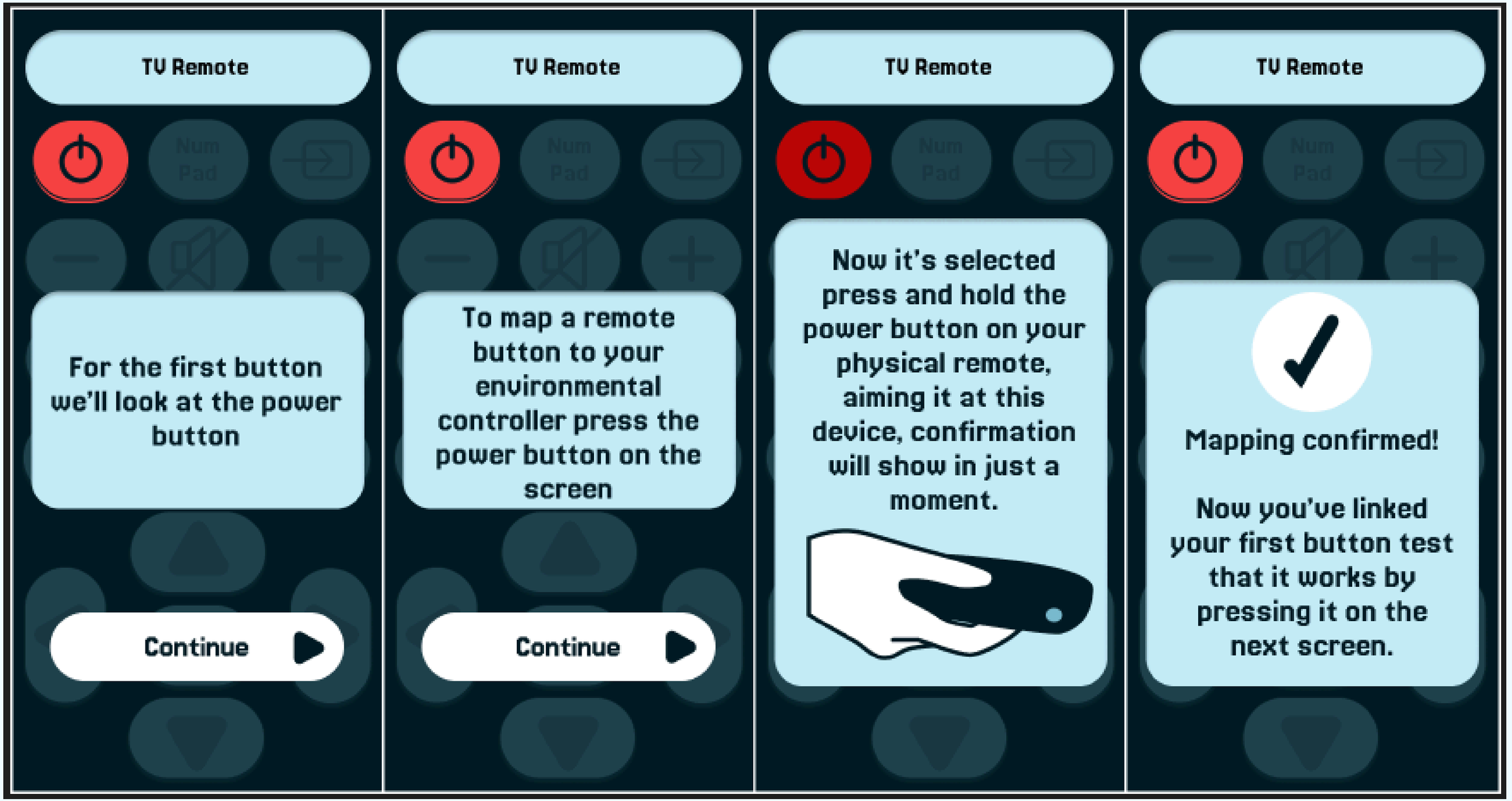
The key goals addressed in the design was explaining how to set up this tech, and giving the user a template, rather than having them select the buttons they wanted one by one by looking at a physical remote. The current programming of this type of environmental control in MND tech layers it behind settings with no prompts for what the user could or should program, putting mental load onto users if they can figure out the capabilities are there in the first place. The design also gives the user ideas of things that can be controlled with IR, such as Aircon, Call Bells, and Lights which aren't as commonly known to all users.

The part I couldn't address through my design aside from some concepts was the adjustability, wanting to have a feature in the settings for users to set custom remotes via a grid pattern from their programmed buttons, this way they can have larger buttons and remove clutter from features they might not need, such as a user who uses their TV for streaming not needing a number pad, ideally this would work as drag and drop for the user to take from a library of their mapped buttons and place them on a grid. As well including some troubleshooting information for users, which I wasn't fully able to do not knowing what some common issues might be.

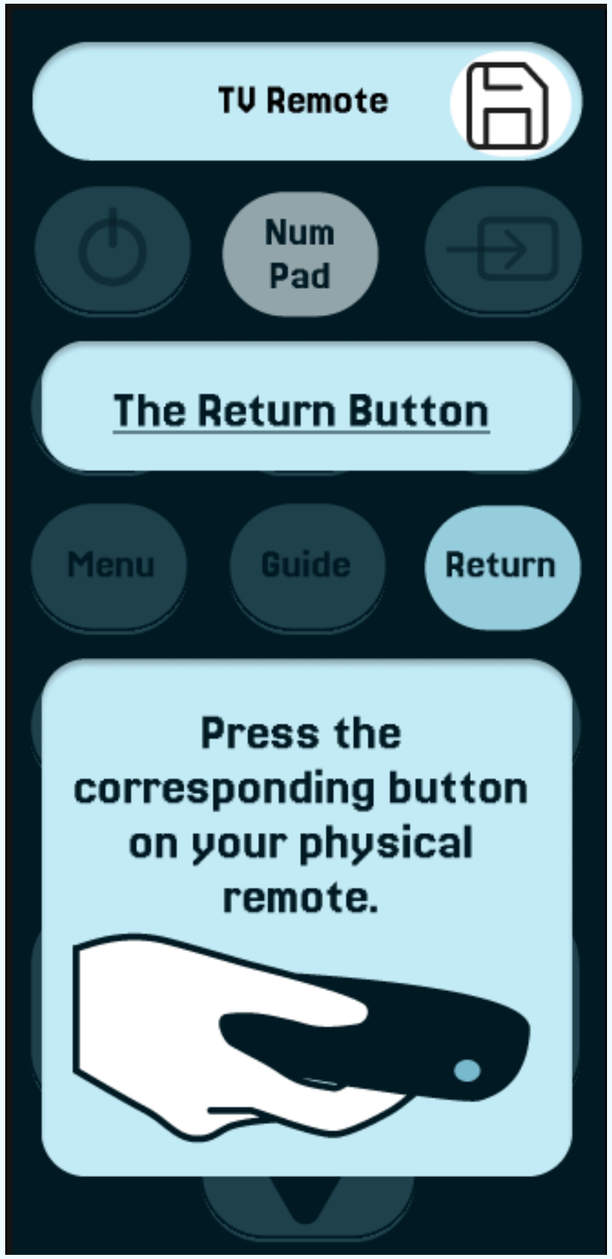
Final Prototype : All Remotes

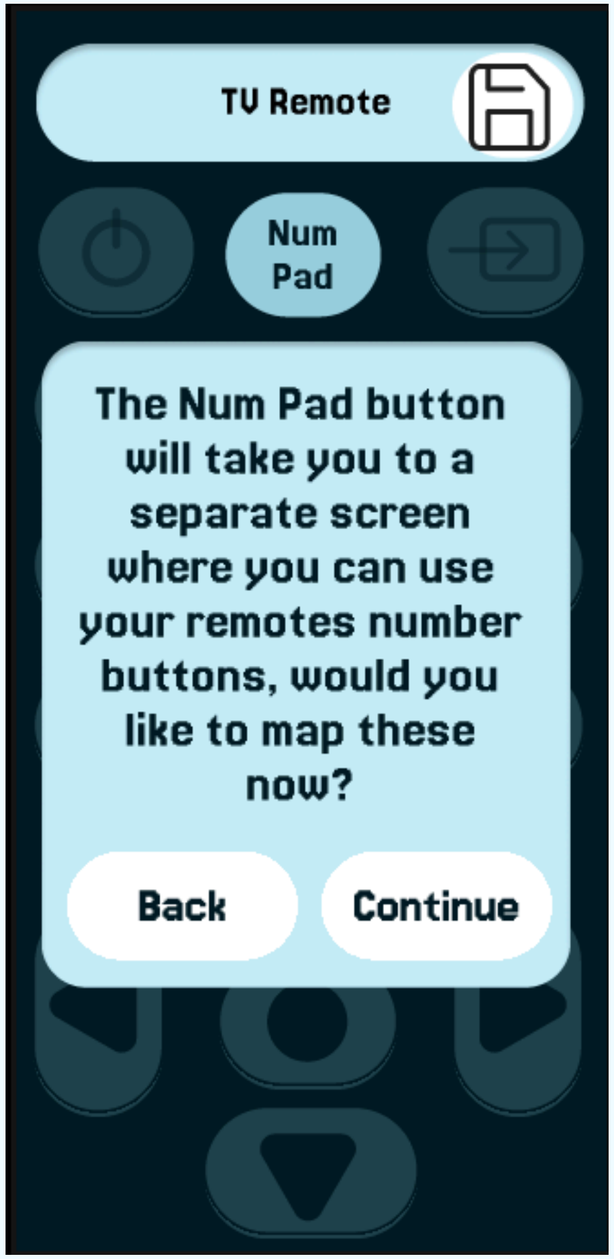


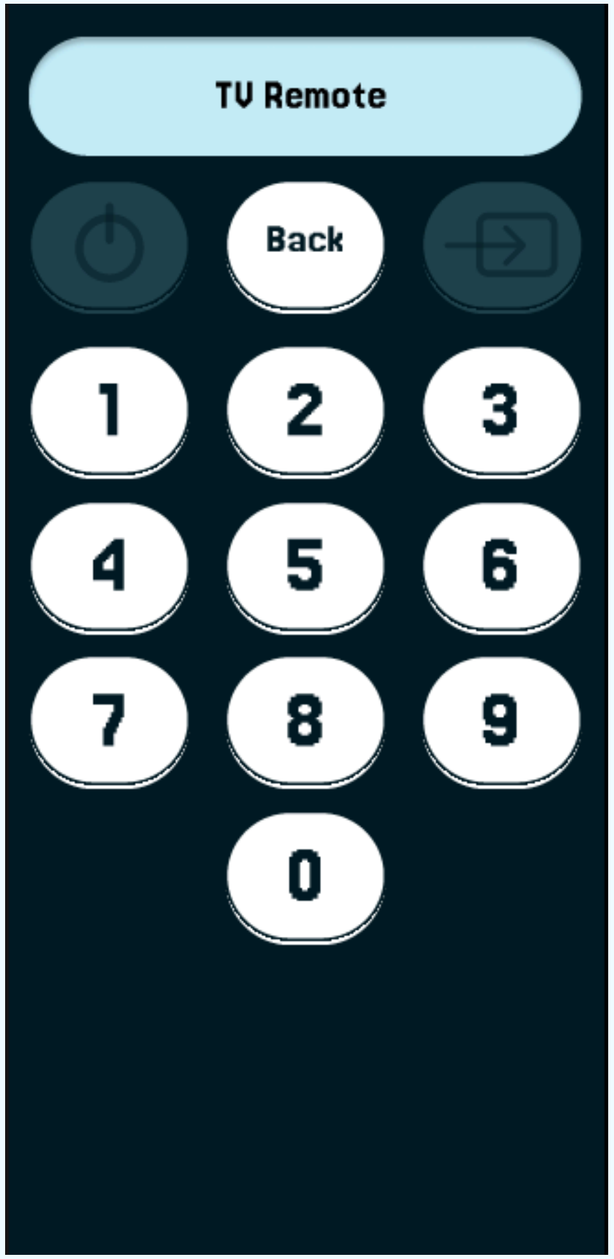
Final Prototype : Onboarding snippet

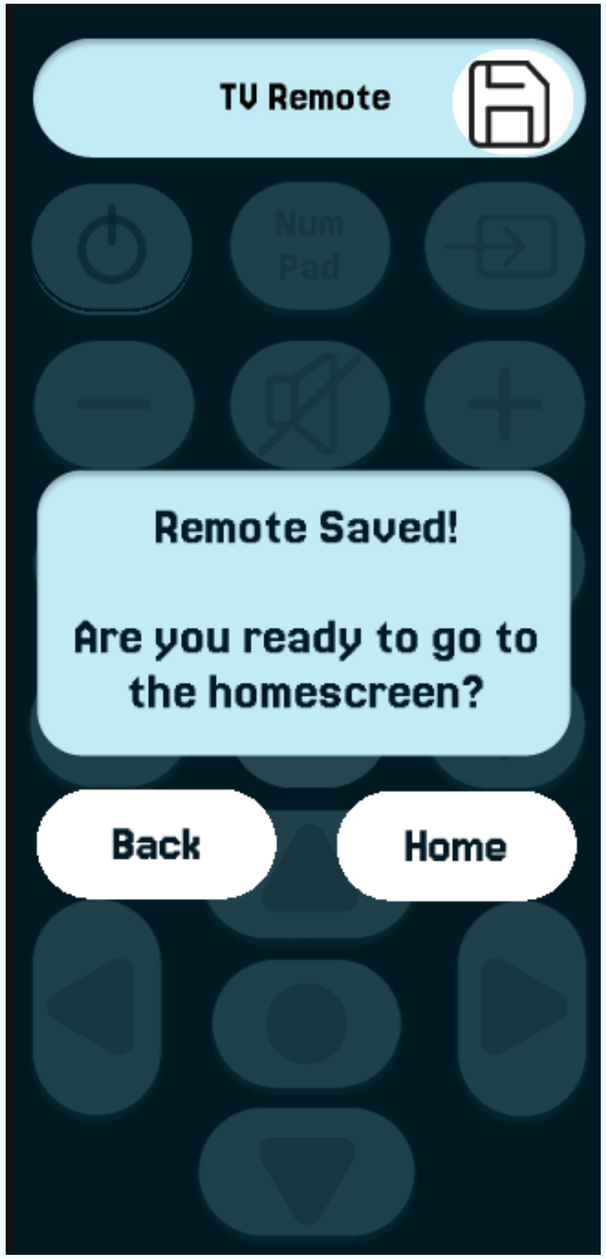


Final Prototype : Onboarding









Final Prototype : Variation on the remotes

